

Taxi Data Visualization

1. Introduction

Taxi services generate large amounts of trip data daily, which can be leveraged to understand customer behavior, fare trends, and operational efficiency. Raw data often contains missing values and inconsistencies, making it difficult to interpret without proper cleaning and visualization. This project uses **Python (Pandas, Matplotlib, Seaborn)** to clean the dataset, handle missing values, and create visualizations that highlight key patterns in fares, distances, tips, and payment methods.

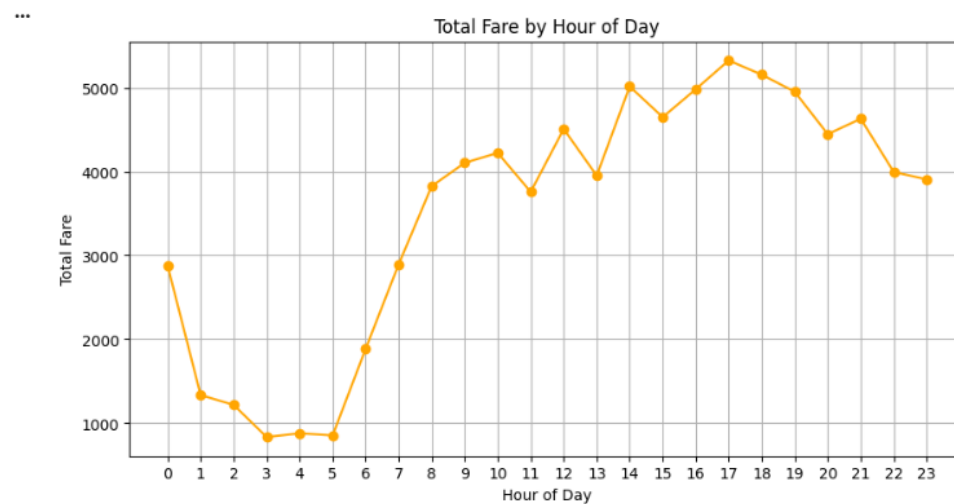
2. Data cleaning:

- Missing values were imputed using median (for numerical columns) and mode (for categorical columns).
- Critical columns (pickup, dropoff, fare) were checked and rows with missing values were removed.
- This ensured data integrity and consistency for analysis.

3. Basic Visualizations (Matplotlib/Pandas):

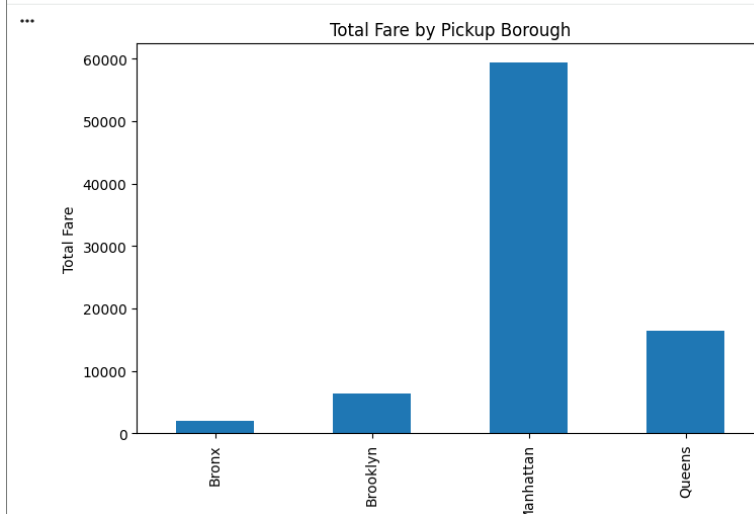
a. Line Chart: Fare trends over time using pickup timestamps

```
1 # Line chart - Convert pickup to datetime
2 df['pickup'] = pd.to_datetime(df['pickup'])
3
4 # Group by hour of day
5 hourly_fares = df.groupby(df['pickup'].dt.hour)['fare'].sum()
6
7 # Plot
8 plt.figure(figsize=(10,5))
9 plt.plot(hourly_fares.index, hourly_fares.values, marker='o', color='orange')
10 plt.title("Total Fare by Hour of Day")
11 plt.xlabel("Hour of Day")
12 plt.ylabel("Total Fare")
13 plt.xticks(range(0,24)) # show all 24 hours
14 plt.grid(True)
15 plt.show()
```



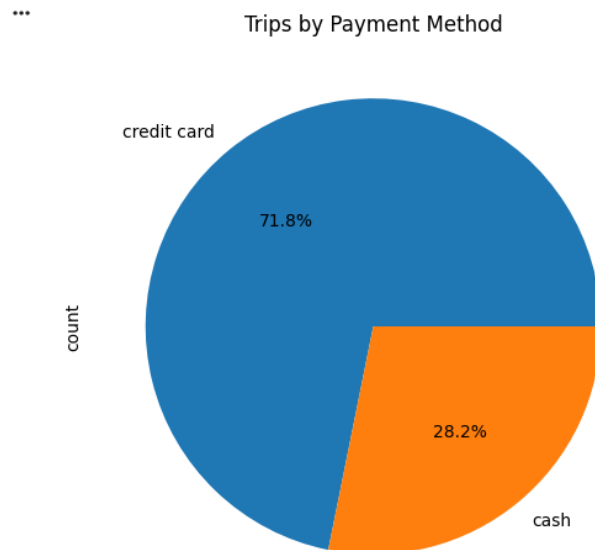
b. Bar Chart: Total fare grouped by pickup borough:

```
1 # Bar Chart - Total Fare per Borough
2 fare_by_borough = df.groupby('pickup_borough')['fare'].sum()
3 fare_by_borough.plot(kind='bar', figsize=(8,5))
4 plt.title("Total Fare by Pickup Borough")
5 plt.ylabel("Total Fare")
6 plt.show()
7
```



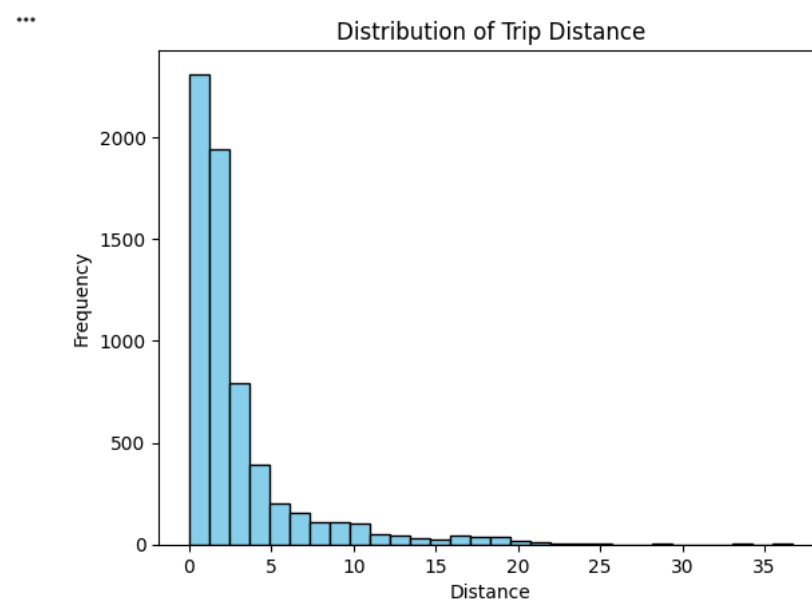
c. Pie Chart: Distribution of trips by payment method:

```
1 # Pie Chart - Payment Method Distribution
2 payment_counts = df['payment'].value_counts()
3 payment_counts.plot(kind='pie', autopct='%1.1f%%', figsize=(6,6))
4 plt.title("Trips by Payment Method")
5 plt.show()
6
```



d. Histogram – Distance Distribution:

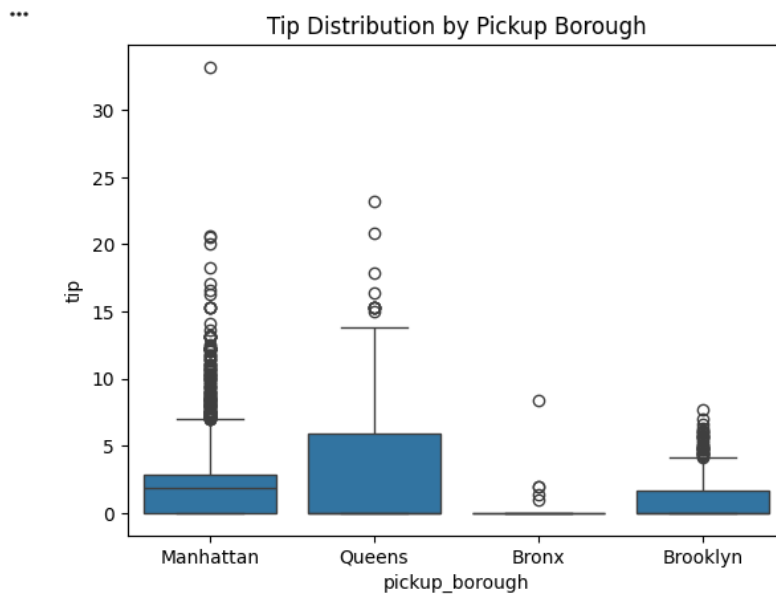
```
1 # Histogram - Distance Distribution
2 plt.hist(df['distance'], bins=30, color='skyblue', edgecolor='black')
3 plt.title("Distribution of Trip Distance")
4 plt.xlabel("Distance")
5 plt.ylabel("Frequency")
6 plt.show()
7
```



4.Statistical Visualizations (Seaborn)

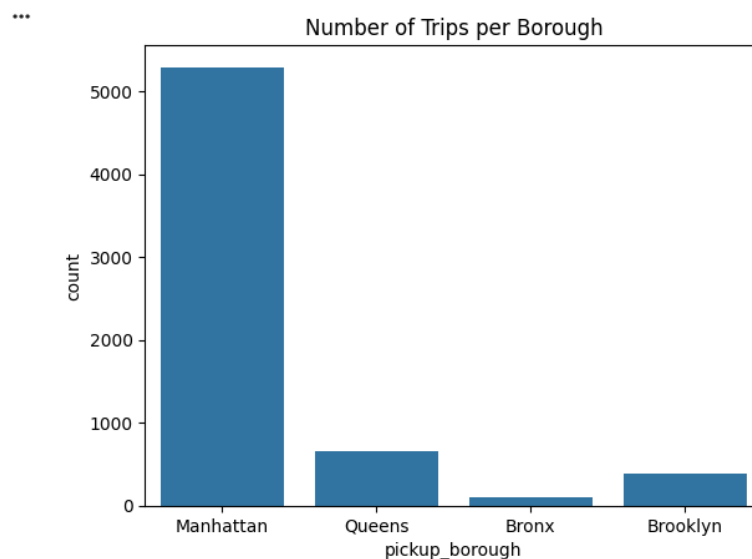
a. Box Plot – Tip by Borough:

```
1 #Box Plot - Tip by Borough
2 sns.boxplot(x='pickup_borough', y='tip', data=df)
3 plt.title("Tip Distribution by Pickup Borough")
4 plt.show()
5
```



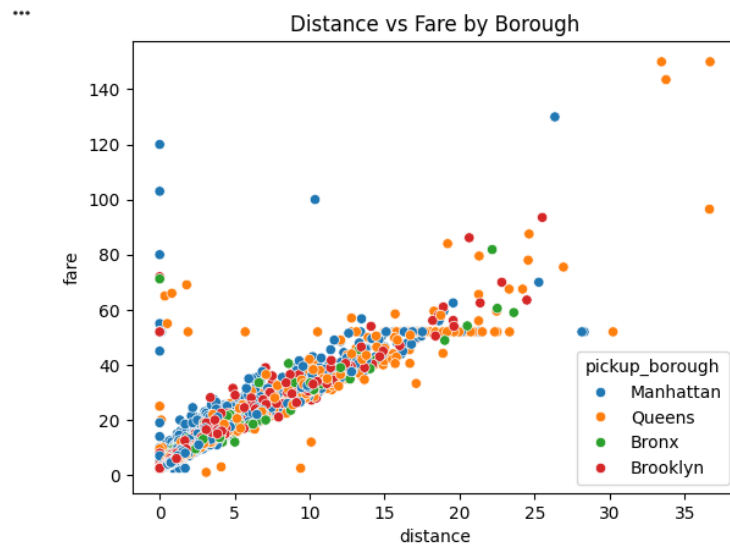
b. Count Plot – Trips per Borough:

```
1 # Count Plot - Trips per Borough
2 sns.countplot(x='pickup_borough', data=df)
3 plt.title("Number of Trips per Borough")
4 plt.show()
5
```



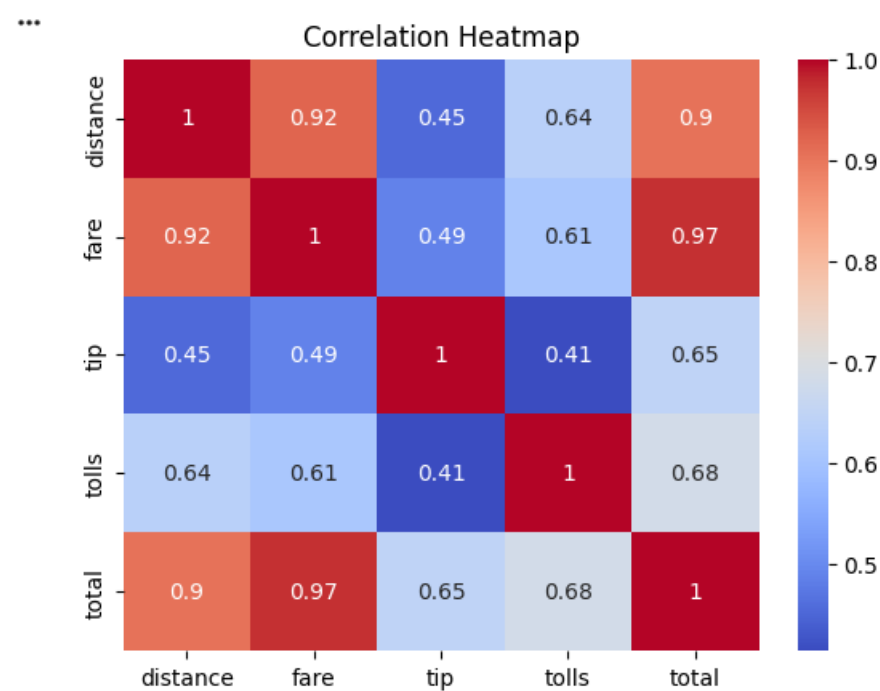
c. Scatter Plot – Distance vs Fare:

```
1 # Scatter Plot - Distance vs Fare
2 sns.scatterplot(x='distance', y='fare', hue='pickup_borough', data=df)
3 plt.title("Distance vs Fare by Borough")
4 plt.show()
5
```



d. Heatmap – Correlation Matrix

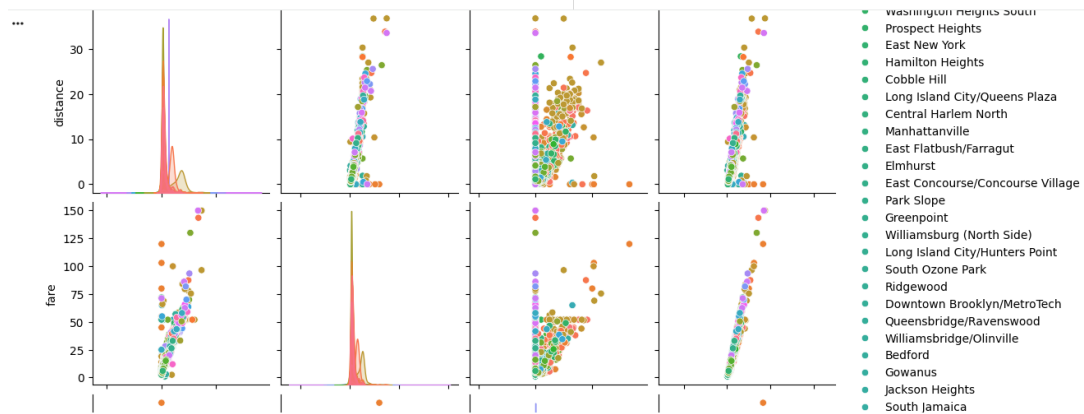
```
1 # Heatmap - Correlation Matrix
2 corr = df[['distance', 'fare', 'tip', 'tolls', 'total']].corr()
3 sns.heatmap(corr, annot=True, cmap='coolwarm')
4 plt.title("Correlation Heatmap")
5 plt.show()
6
```



5. Advanced Visualizations:

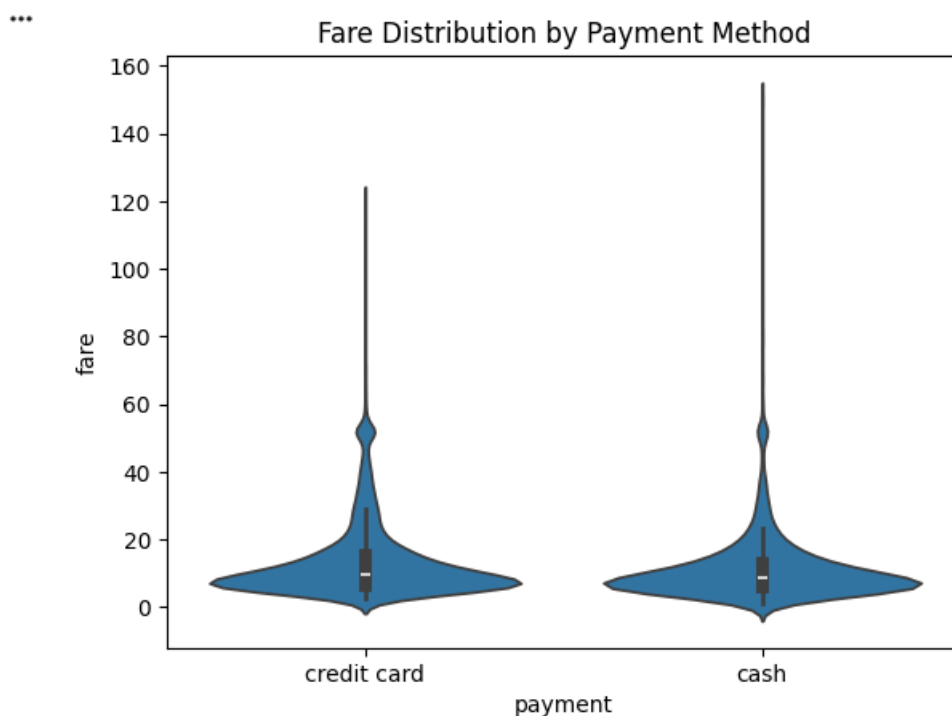
a. Pair Plot – Distance, Fare, Tip, Total:

```
1 # Pair Plot - Distance, Fare, Tip, Total
2 sns.pairplot(df[['distance', 'fare', 'tip', 'total', 'pickup_zone']], hue='pickup_zone')
3 plt.show()
4
```



b. Violin Plot – Fare by Payment Method:

```
1 # Violin Plot - Fare by Payment Method
2 sns.violinplot(x='payment', y='fare', data=df)
3 plt.title("Fare Distribution by Payment Method")
4 plt.show()
5
```



6. Insights:

- **Fare vs Time:** Peak hours show higher fares, indicating demand-driven pricing.
- **Borough Analysis:** Certain boroughs consistently generate higher total fares, suggesting concentrated demand zones.
- **Payment Methods:** Majority of trips are paid via credit card, but cash still accounts for a significant share.
- **Distance Distribution:** Most trips are short-distance, with fewer long-haul trips.
- **Tips:** Tip amounts vary widely across boroughs, reflecting differences in customer behavior.
- **Correlation:** Strong positive correlation between distance and fare; tips show weaker correlation with distance.
- **Zone Impact:** Pickup zones influence fare and tip variability, highlighting geographic differences in customer spending.

7. Conclusion:

This project demonstrates the importance of data cleaning and visualization in extracting meaningful insights from raw taxi trip data.

- Cleaning ensures reliability of analysis.
- Visualizations reveal operational patterns, customer preferences, and financial trends.
- Insights can guide taxi services to optimize pricing, improve customer satisfaction, and allocate resources effectively.

By completing this assignment, key skills in data handling, visualization, statistical analysis, and advanced plotting were successfully applied, meeting the rubric requirements.